

Final Report

From Deterministic Rate Equations to Stochastic Optical Readout of Nitrogen-Vacancy Centres

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1 Introduction

1.1 Nitrogen-vacancy centres in diamond

The nitrogen-vacancy (NV) centre is a defect in diamond, formed by substituting a carbon atom with nitrogen atom inside the lattice, leaving an adjacent site vacant. Given they're optical accessibility, long lived spin states, and operability at room temperature they have become an important platform for quantum sensing, nanoscale imaging, and studies of solid-state spin physics [1–3]. The defect can be in several charge states, with additional spin states relevant for the NV^-charge state. The neutral charge state NV^0remains important because optical excitation can allow populations to transfer between these charge configurations [4].

The electronic ground state of NV^-has spin projections of $m_s = 0$ and $m_s = \pm 1$. Even in the absence of a magnetic field, the $m_s = 0$ level is separated from the $m_s = \pm 1$ state by approximately 2.87 GHz [1]. These spin sub-levels can be coherently manipulated using microwaves and detected optically.

1.2 Spin-dependent optical cycling and readout

NV centres under green illumination promote the NV^-triplet ground state into an excited triplet state. As these excited states relax back to the NV^-ground state, they emit a photon and thus can be considered radiative transitions. These radiative pathways approximately preserve spin projection; however, there exists a metastable singlet state that allows for non-radiative transfers. This non-radiative transition is considerably more probable for the excited $m_s = \pm 1$ states than for the excited $m_s = 0$ state [1, 5]. As a consequence, NV centres initialised in $m_s = 0$ will emit more photons near the beginning of an optical pulse than one prepared in $m_s = \pm 1$. Additionally, the decay from the metastable state prefers to return the population to the $m_s = 0$ ground state, so continued illumination also polarises the spin.

These processes provide both spin initialisation and spin readout, but they also impose a fundamental trade-off. A very short observation window contains strong spin contrast with few emitted photons. A longer window will collect more light, however the optical polarisation progressively erases the difference between the initial spin states. The choice of readout duration is therefore a quantitative optimisation problem rather than an arbitrary experimental setting.

This system is represented by a seven-state population model. Since an external magnetic field is not applied, the $m_s = \pm 1$ spin states can be reduced to just $m_s = -1$. Four states describe the ground and excited triplet levels associated with NV^-, alongside an additional metastable state. Two states represent the ground and excited configurations of NV^0. Refer to later sections for more clarity.

1.3 Computational motivation

The state populations obey a coupled system of linear rate equations. For a constant laser intensity, propagation by a matrix exponential provides a deterministic model of the time-dependent populations and mean photoluminescence (PL). This description is useful for checking probability conservation, comparing characteristic transition times, and calculating the expected number of photons collected within a certain readout window. It however, does not represent shot-to-shot fluctuations of a discrete photon-counting experiment. A Poisson model provides the natural description for these discrete photon events.

This progression; from rate equations, to photon statistics, to parameter inference, also provides a useful computational test case. It permits the numerical method to be validated against physical constraints.

1.4 Aim, research questions, and scope

The central aim of this project is to determine how accurately and efficiently a seven-state rate-equation model can describe spin-dependent NV centre photoluminescence, and how finite photon statistics constrain both optical readout and recovery of transition rates. This aim is addressed primarily through four research questions:

1. How do the state populations and spin-dependent PL signals evolve during constant optical excitation?
2. What readout duration gives the best separation between the expected signals from initial $m_s = 0$ and $m_s = -1$ states?
3. How do Poisson photon statistics alter the apparent separation of the two spin states and the performance of a count threshold?
4. Can synthetic time-resolved photon counts be used to recover the laser-excitation parameter β and the spin-dependent transition rates k_{35} and k_{45}?

This investigation deliberately avoids modelling a microwave Hamiltonian, coherent spin dynamics, magnetic-field-dependent spectra, spatial charge diffusion, or every microscopic singlet level. The purpose of this project is to test identifiability and numerical methodology under controlled conditions.

The remainder of this report first defines the seven-state physical model and its assumptions. It then describes deterministic propagation, photon-count simulation, readout optimisation and parameter inference. The results are presented in the same order before the physical limitations, numerical performance and possible extensions are discussed.

2 Physical Model

2.1 Seven-state energy-level structure

The optical cycle was represented by the seven-state model shown in Figure 1. The ordering of the states is fixed throughout the code and is listed in Table 1. No coherent superposition of spin states is modelled.

The spin dependence enters principally through the intersystem crossing rates k_{35} and k_{45}. Since $k_{45} > k_{35}$, population beginning in the $m_s = -1$ excited state is shelved into the non-radiative singlet pathway more quickly than population beginning in $m_s = 0$. The $m_s = 0$ state is therefore brighter immediately after the laser is turned on.

States $|6\rangle$ and $|7\rangle$ incorporate photo-ionisation from NV^- and recombination from NV^0. Their inclusion is important because charge conversion affects both the available NV^- population and the detected light during a readout pulse [6, 7].

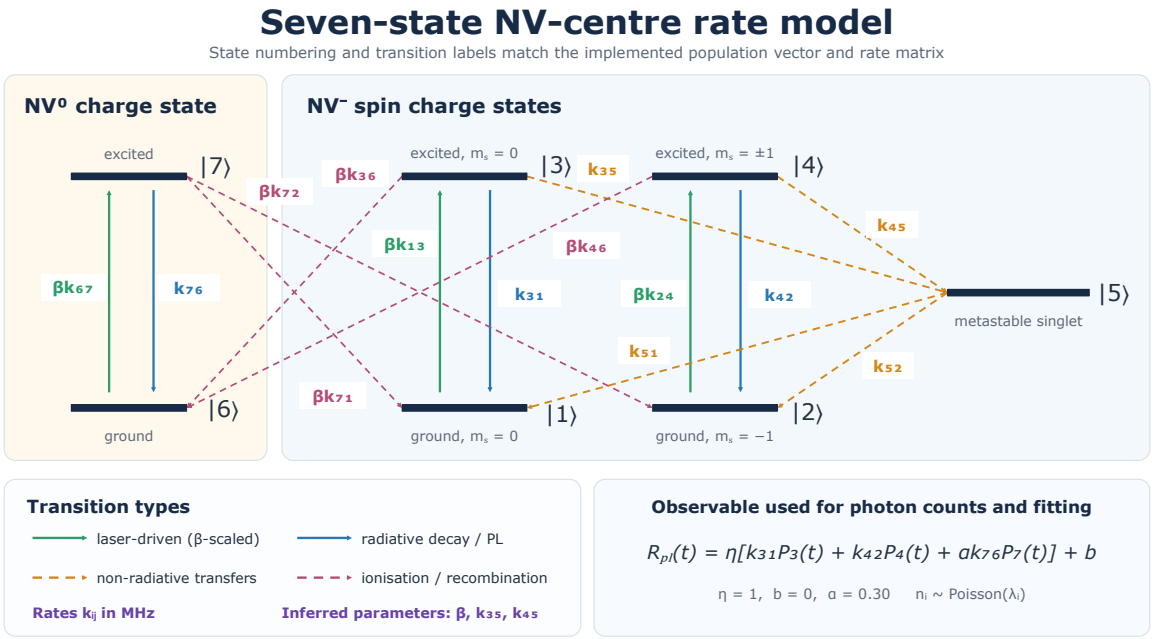


Figure 1: Seven-state model implemented in this project. Green arrows are laser-driven excitation, blue arrows are radiative decay, orange dashed arrows are metastable pathways, and purple dashed arrows are ionisation or recombination. The state numbering, transition topology and rate constants follow the model used by Wang [6]. The photoluminescence equation and the fitted parameter subset are included to connect the physical model to the subsequent simulations.

Table 1: State ordering used in the population vector and in every row and column of the transition matrix.

State	Charge state and manifold	Spin label
$\|1\rangle$	NV^- triplet ground state	$m_s = 0$
$\|2\rangle$	NV^- triplet ground state	$m_s = -1$
$\|3\rangle$	NV^- triplet excited state	$m_s = 0$
$\|4\rangle$	NV^- triplet excited state	$m_s = -1$
$\|5\rangle$	metastable singlet state	—
$\|6\rangle$	NV^0 ground state	—
$\|7\rangle$	NV^0 excited state	—

2.2 Rate-equation model

Let

$$\mathbf{P}(t) = \begin{bmatrix} P_1(t) & P_2(t) & P_3(t) & P_4(t) & P_5(t) & P_6(t) & P_7(t) \end{bmatrix}^{\mathsf{T}} \tag{1}$$

contain the population of each state. Time is measured in microseconds and all tabulated rates are in megahertz, numerically equivalent to μs^{-1}. The populations obey the linear master equation

$$\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t} = \mathbf{A}(\beta)\mathbf{P}, \tag{2}$$

where A_{ij} is the rate of transfer from state j into state i. Using the same directional rate labels as the Python implementation, the matrix defined by `rate_matrix` is

$$\mathbf{A}(\beta) = \begin{bmatrix} -\beta k_{13} & 0 & k_{31} & 0 & k_{51} & 0 & \beta k_{71} \\ 0 & -\beta k_{24} & 0 & k_{42} & k_{52} & 0 & \beta k_{72} \\ \beta k_{13} & 0 & -(k_{31}+\beta k_{36}+k_{35}) & 0 & 0 & 0 & 0 \\ 0 & \beta k_{24} & 0 & -(k_{42}+k_{45}+\beta k_{46}) & 0 & 0 & 0 \\ 0 & 0 & k_{35} & k_{45} & -(k_{51}+k_{52}) & 0 & 0 \\ 0 & 0 & \beta k_{36} & \beta k_{46} & 0 & -\beta k_{67} & k_{76} \\ 0 & 0 & 0 & 0 & 0 & \beta k_{67} & -(k_{76}+\beta k_{71}+\beta k_{72}) \end{bmatrix}. \tag{3}$$

Every off-diagonal entry is non-negative and each diagonal entry is the negative sum of the rates leaving that state. Consequently the columns of $\mathbf{A}$ sum to zero, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{i=1}^{7} P_i(t) = \mathbf{1}^{\mathsf{T}}\mathbf{A}\mathbf{P} = 0. \tag{4}$$

This gives a direct physical and numerical check: a normalised initial state must remain normalised and non-negative. For constant β, the exact finite-time solution is

$$\mathbf{P}(t) = \exp[\mathbf{A}(\beta)t]\,\mathbf{P}(0), \tag{5}$$

which is evaluated using SciPy's matrix exponential. The project also uses an adaptive Runge–Kutta solver for time-dependent laser sequences.

Two initial spin preparations are compared. Both assume an initial charge mixture of 70% NV^- and 30% NV^0, with no excited-state or singlet population:

$$\begin{aligned} \mathbf{P}_{m_s=0}(0) &= (0.7, 0, 0, 0, 0, 0.3, 0)^{\mathsf{T}}, \\ \mathbf{P}_{m_s=-1}(0) &= (0, 0.7, 0, 0, 0, 0.3, 0)^{\mathsf{T}}. \end{aligned} \tag{6}$$

This charge ratio is an explicit modelling choice for the simulations rather than a universal material constant.

2.3 Laser-dependent transitions and rate constants

The dimensionless parameter $\beta \geq 0$ scales transitions driven by the green laser. Thus optical excitation, ionisation and recombination appear as βk, whereas spontaneous radiative decay and intersystem crossing are not multiplied by β. The reference simulations use $\beta = 0.7$, and the parameter-inference study later treats it as unknown.

The numerical constants used in the simulations are listed explicitly in Table 2. Their values were taken from Wang's thesis [6].

The large ratio $k_{45}/k_{35} = 5.5$ is the main source ofspin contrast. By contrast, the ratio $k_{51}/k_{52} \approx 4.67$ favours return to $m_s = 0$ and therefore causes the two initial conditions to lose their memory as optical pumping proceeds.

Table 2: Directional transition-rate constants used in the simulations. All rates are in MHz (μs^{-1}).

Symbol	Transition	Value	Term in $\mathbf{A}$
k_{13}	$\|1\rangle \to \|3\rangle$ optical excitation	65	βk_{13}
k_{24}	$\|2\rangle \to \|4\rangle$ optical excitation	65	βk_{24}
k_{31}	$\|3\rangle \to \|1\rangle$ radiative decay	65	k_{31}
k_{42}	$\|4\rangle \to \|2\rangle$ radiative decay	65	k_{42}
k_{35}	$\|3\rangle \to \|5\rangle$ intersystem crossing	10	k_{35}
k_{45}	$\|4\rangle \to \|5\rangle$ intersystem crossing	55	k_{45}
k_{51}	$\|5\rangle \to \|1\rangle$ singlet decay	7	k_{51}
k_{52}	$\|5\rangle \to \|2\rangle$ singlet decay	1.5	k_{52}
k_{36}	$\|3\rangle \to \|6\rangle$ ionisation	10	βk_{36}
k_{46}	$\|4\rangle \to \|6\rangle$ ionisation	10	βk_{46}
k_{67}	$\|6\rangle \to \|7\rangle$ optical excitation	40	βk_{67}
k_{76}	$\|7\rangle \to \|6\rangle$ radiative decay	40	k_{76}
k_{71}	$\|7\rangle \to \|1\rangle$ recombination	100/3	βk_{71}
k_{72}	$\|7\rangle \to \|2\rangle$ recombination	200/3	βk_{72}

2.4 Photoluminescence and photon-detection model

Only radiative decays contribute directly to the modelled photoluminescence (PL). The expected detected rate is

$$R(t) = \eta \left[k_{31} P_3(t) + k_{42} P_4(t) + \alpha k_{76} P_7(t)\right] + b, \tag{7}$$

where η is an overall detection scale, b is a constant background and α is the relative collection of NV^0 emission. Wang's experiment optically filtered the neutral-state emission and retained the fraction $\alpha = 0.30$ [6]; the same value is used here. The present simulations set $\eta = 1$ and $b = 0$. Their PL axis is treated as a normalised model rate rather than a calibrated laboratory count rate. This convention preserves the dynamics and relative contrast, but absolute comparison with an experiment would require detector efficiency, optical losses and background to be measured.

For a readout interval $[0, T]$, the mean number of detected photons is

$$\mu(T) = \int_0^T R(t)\, \mathrm{d}t. \tag{8}$$

The deterministic calculation returns μ, which is not restricted to only integers. Individual simulated measurements instead draw

$$N \sim \mathrm{Poisson}(\mu)\,, \qquad \mathrm{Var}(N) = \mu. \tag{9}$$

This separation between the mean optical model and discrete detected counts is central to the later readout and parameter-inference calculations.

2.5 Verification against the reference model

Verification was carried out at three levels. Firstly, all eight independent rate values in Equations (3) and Table 2 were checked against Wang's seven-state model and rate table [6]. Wang reports that this model gives excellent agreement with actual experiment data (refer to figure 3.4 in his paper). The present work reproduces the published model.

Second, structural tests were applied directly to the implemented matrix: self-consistency checks. Third, constant-laser trajectories from the matrix exponential were compared with the

independent numerical solver implementation on $0 \leq t \leq 1\,\mu$s using 1001 output times. The largest population difference across both initial spin states was 7.2×10^{-13}, while the maximum deviation of total population from unity was 1.1×10^{-14}.

These tests demonstrate internal numerical consistency; they do not remove the physical approximations of the model.

2.6 Computational workflow

Figure 2 shows how the forward model supports the four investigations in the project. Keeping constants, state propagation, and PL calculation in one base module avoids having to duplicate the physical model.

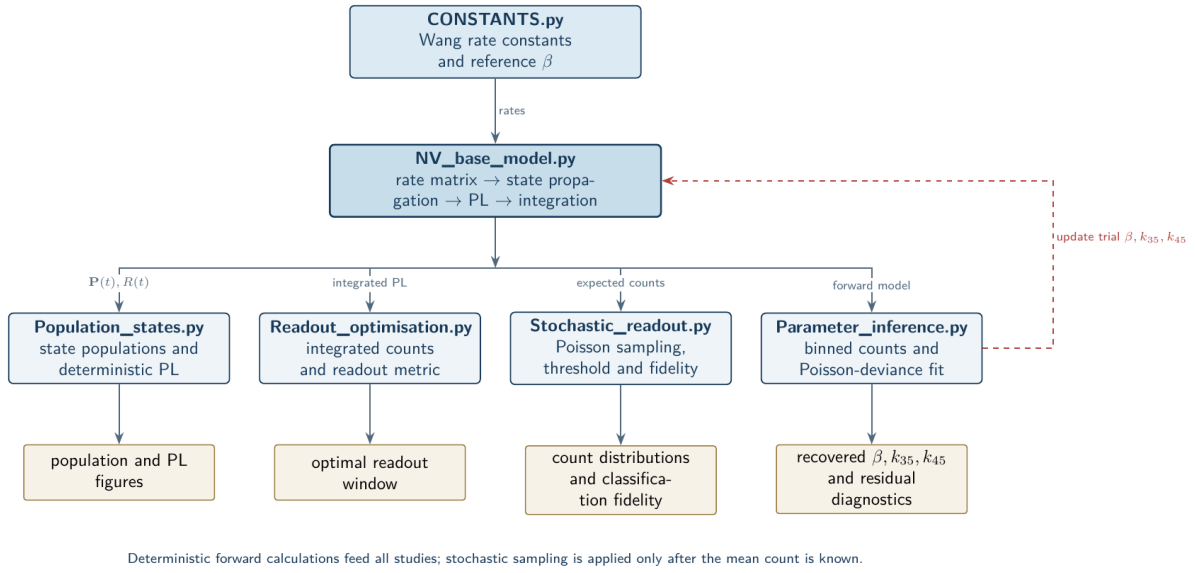


Figure 2: Workflow and dependencies of the project code.

The workflow also separates deterministic and stochastic quantities. The base model produces expected populations and photon rates. Numerical integration converts the rate into an expected count, and only then is a Poisson random draw used to represent a measurement.

3 Computational Methods

3.1 Integrated photoluminescence and readout optimisation

The instantaneous PL rate is evaluated using Equation (7). The cumulated expected count for both spin preparations at time t_j is then obtained with the trapezoidal rule:

$$\mu_s(t_j) \approx \sum_{i=0}^{j-1} \frac{R_s(t_i) + R_s(t_{i+1})}{2} (t_{i+1} - t_i)\,. \tag{10}$$

Considering independent Poisson counts, the variance of the difference between the two measurements is the sum of their means. The signal-to-noise ratio is therefore defined as the mean signal

divided be the standard deviation,

$$\mathrm{SNR}(T) = \frac{|\mu_0(T) - \mu_{-1}(T)|}{\sqrt{\mu_0(T) + \mu_{-1}(T)}}. \tag{11}$$

Equation (11) is evaluated at every point of the 1 ns scale and the reported readout duration is

$$T^\star = \arg\max_T \mathrm{SNR}(T). \tag{12}$$

The value at $T = 0$ is assigned an SNR (signal-to-noise ratio) of zero to avoid dividing by zero.

3.2 Stochastic photon-counting simulation

The deterministic model yields a mean count but not individual measurements. To construct repeated readouts, independent M samples are drawn for each spin initialisation according to

$$n_s^{(q)} \sim \mathrm{Poisson}[\mu_s(T^\star)], \qquad q = 1, \ldots, M. \tag{13}$$

A NumPy random number generator with seed 676767 was used, and $M = 10,000$. As numerical check on the simulation, the sample mean and variance were compared with the deterministic Poisson mean $\mu_s(T^\star)$.

Because the $m_s = 0$ initialisation produced a larger mean count, and integer threshold c allows us to assign the spin of single shots according to

$$spin(n; c) = \begin{cases} 0, & n \geq c, \\ -1, & n < c. \end{cases} \tag{14}$$

The distributions are displayed as discrete probability masses using side by side bars at each integer photon count.

3.3 Parameter inference

The inverse problem estimates

$$\boldsymbol{\theta} = (\beta, k_{35}, k_{45})^\mathsf{T}, \tag{15}$$

while all other rates retain the values in Table 2. These parameters were chosen because β controls the laser driven processes, and k_{35} and k_{45} control the spin-dependent transfer into the singlet metastable state. Synthesised data are used so that the fitted values can be compared with a known parameter without the need for experimental calibration.

For each initial spin state, the interval $0 \leq t \leq 0.6\,\mu$s is split into 120 bins of equal width. The expected count in bin j, accumulated over 5000 repetitions is approximated at the bin centre t_j^c,

$$\nu_j = 5000 * \Delta t_{\mathrm{bin}}\, R(t_j^c). \tag{16}$$

Synthetic observations in each bin is independently sampled as $y_j = Poisson(\nu_j)$, using the same fixed seed as the stochastic readout calculation.

The Poisson variance changed with the expected count, thus each observation contributes the signed square root Poisson deviance residual [8],

$$r_j = \mathrm{sign}(y_j - \nu_j)\sqrt{2\left[\nu_j - y_j + y_j \ln\left(\frac{y_j}{\nu_j}\right)\right]}. \tag{17}$$

For $y_i = 0$, the log term is defined to be zero for continuity. Expected values are bounded below by 10^{-12}, and the deviance is cut off at 0 before taking the square root.

By utilising a log transformation, $\mathbf{q} = \ln \boldsymbol{\theta}$, this ensures only positive values. Then, implement SciPy non-linear least squares solver,

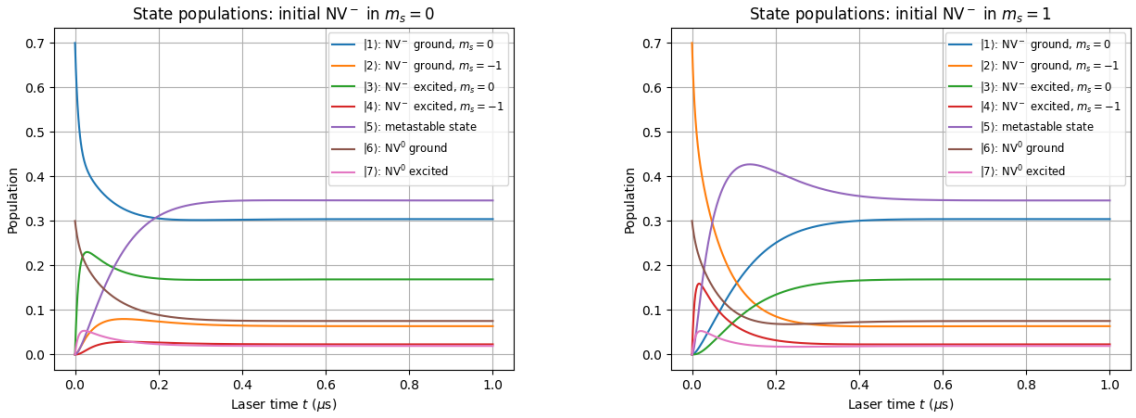
$$\mathbf{q}^\star = \arg\min \frac{1}{2} \sum_{i=1}^{240} r_i(\exp \mathbf{q})^2 \tag{18}$$

This is summed over both spin states, where the first 120 bins are $ms = 0$ initialised and the last 120 bins are $ms = \pm 1$ initialised. The starting point is $(0.5, 7, 40)^\mathsf{T}$, deliberately displaced from the true parameters. The lower bounds are $(0.03, 1, 5)^\mathsf{T}$ and the upper bounds are $(3, 50, 150)^\mathsf{T}$.

4 Results and Analysis

4.1 Population dynamics and deterministic photoluminescence

Figure 3 shows the seven populations for the two initial spin preparation. At $t = 0$, the NV^-population is entirely in either $|1\rangle$ or $|2\rangle$, and the initial 30% NV^0component occupies $|6\rangle$. laser excitation first transfers population into the corresponding triplet state. The $m_s = -1$ preparation then produces a substantially larger singlet metastable population because $k_{45} = 55\,\text{MHz}$ is greater than $k_{35} = 10\,\text{MHz}$. Population leaving this singlet is preferentially returned to the $m_s = 0$ ground state, so the initially different trajectories approach the same laser driven distribution at later times. This convergence shows optical spin polarisation.



(a) Initial $m_s = 0$.

(b) Initial $m_s = -1$.

Figure 3: Simulated population dynamics under constant green illumination. Both preparations begin with 70% NV^- and 30% NV^0. Both initialisations converge towards the same long-time distribution.

The total population remains normalised throughout both simulations. Largest deviations were of order of magnitude of 10^{-14}, which can be associated with floating point round off error.

The corresponding PL races are shown in Figure 4. Both signals rise rapidly as the excited states become populated. The $m_s = 0$ prepared state reaches the larger early time PL because less of its excited population is removed by the non-radiative intersystem crossing. The $m_s = -1$ prepared state is suppressed by the rapid transfer into the metastable state. As optical pumping

removes the memory of their initialised spin state, the two PL curves approach the same value. The spin-dependent information is therefore concentrated at the beginning of the laser pulse.

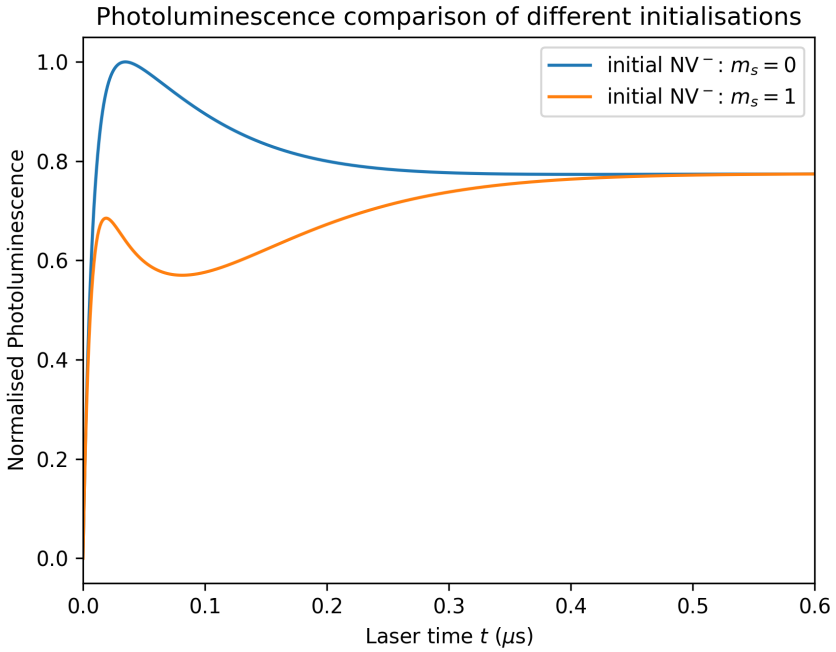


Figure 4: Deterministic PL for the two initial spin preparations, normalised. The early-time separation is caused by spin-dependent intersystem crossing, while the later convergence follows from optical pumping. Figure matches exactly with Di Wang's, page 66 figure 3.4.

Integrating over the compelete time interval gives an expected count of 13.005622 for initial $m_s = 0$ and 11.973622 for initial $m_s = -1$. These are mean counts, and are therefore not restricted to integer values.

4.2 Optimisation of the readout window

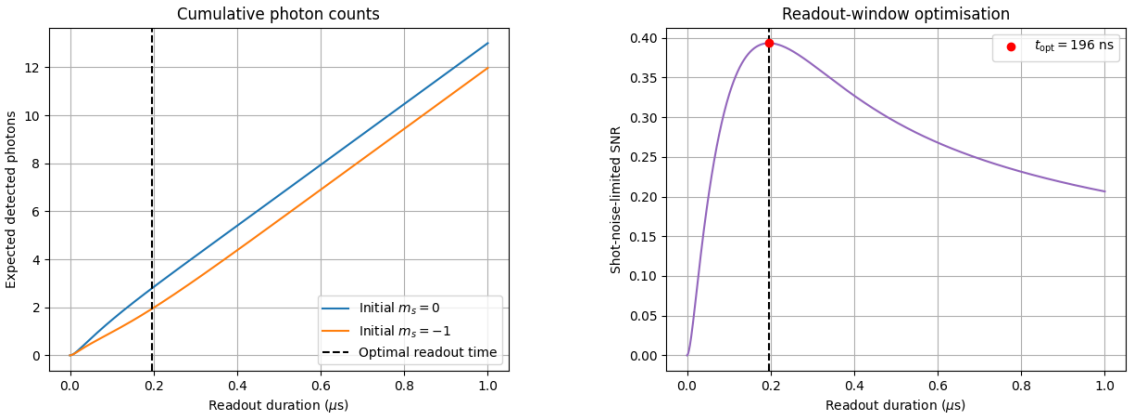
Figure 5 compares the accumulated counts with the shot noise limited SNR. The cumulative counts separate quickly after laser turn on, but the slops become similar once the two PL rates converge. Extending the readout then adds photons to both measurements without preserving the same relative spin constrast.

The optimal readout time occurs at $T^\star = 0.196\mu$s = 196ns. At this duration, the expected counts are 2.791968 for initial $m_s = 0$ and 1.936564 for initial $m_s = -1$, giving a single shot SNR of 0.393376. This readout window is exactly as anticipated in the introduction: very short windows are photon limited, while longer windows retain increasingly less information about the initial spin.

4.3 Simulated photon count distributions

The deterministic means at $T^\star$ were used to generate 10,000 Poisson distributed readouts for both initial states. From Table 3, empirical mean and variance are close to on another and to the prescribed deterministic mean, which is to be expected for the implemented Poisson model.

Despite their different means, the distributions in Figure 6 overlap considerably. Photon counts



(a) Cumulative expected counts.

(b) Shot-noise-limited SNR.

Figure 5: Readout window optimisation.

Table 3: Comparison between deterministic and simulated photon-count statistics at the optimised readout duration.

Initial state	Deterministic mean	Sample mean	Sample variance
$m_s = 0$	2.7920	2.7689	2.7608
$m_s = -1$	1.9366	1.9402	1.9144

of one, two and three occur frequently for both prepared states, so no integer threshold perfectly separates these states. The selected threshold is $c = 3$. Observations of zero, one or two photons are assigned to $m_s = -1$. Observations of three or more photons are assigned to $m_s = 0$. The estimated classification accuracy is 0.6053. The stochastic result shows that while the two mean signals are distinct, their discrete single readout counts have strongly overlapping distributions.

4.4 Recover of model parameters

The final investigation fitted β, k_{35}, and k_{45} to the artificial counts. The optimisation sequence terminated successfully after function evaluations. Table 4 compares the recovered values with the parameters used to generate the artificial experiment. Each fitted value is close to its corresponding true value, and remains well inside the boundary restrictions.

Table 4: Generating and fitted parameters for the synthetic-data inference calculation.

Parameter	Generating value	Fitted value
β	0.7000	0.6886
k_{35}	10.0000	9.6535
k_{45}	55.0000	53.2065

The upper graph of Figure 7 shows that the fitted curves reproduce both the rapid spin contrast and slower convergence. The fitted solid curves lie close to the dashed deterministic curve. The simulated points fluctuate around them, a byproduct of Poisson counting noise.

The lower graph of Figure 7 shows no obvious time dependent pattern across the full time ranges for both preparations. Together, the parameter values and the residual plot show that the

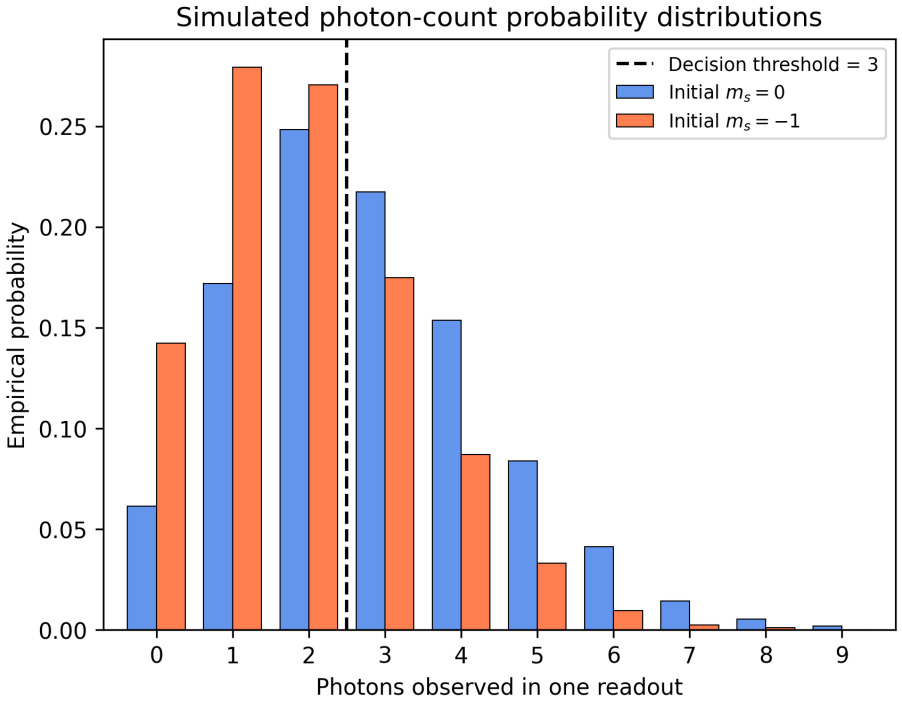


Figure 6: Empirical photon count probabilities from 10,000 simulated readouts of each initial state at $T^\star$. The vertical line at 2.5 represents the integer decision threshold $c = 3$.

implemented forward model can recover the generating rates from this reproducible synthetic data set.

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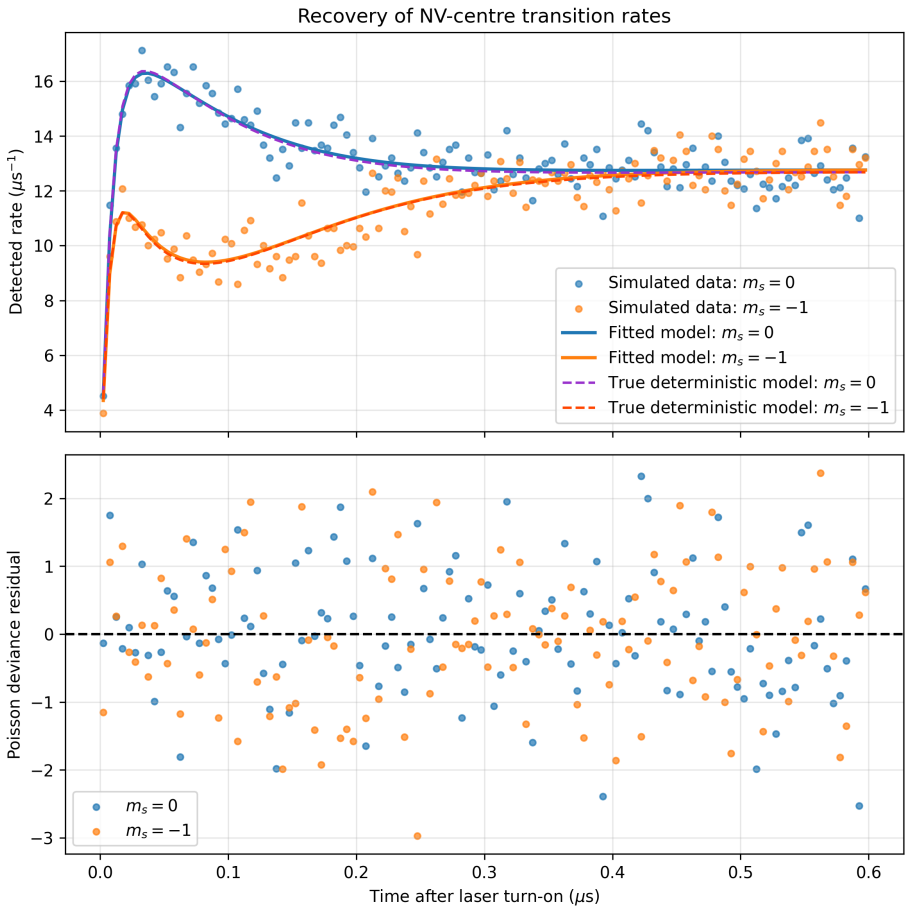


Figure 7: Parameter-inference result. The upper graph compares the simulated time-binned rates with the fitted and generating deterministic curves. The lower graph shows the signed Poisson deviance residual for every time bin.

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